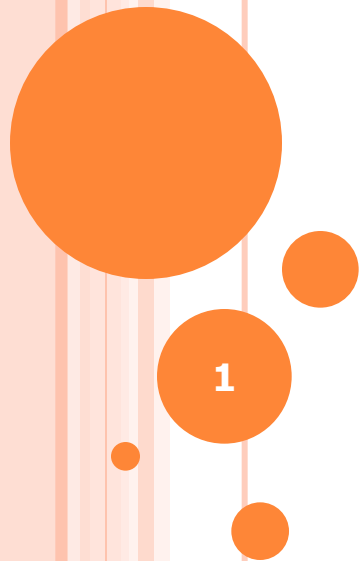


DEVELOPMENT OF MUSCULAR SYSTEM

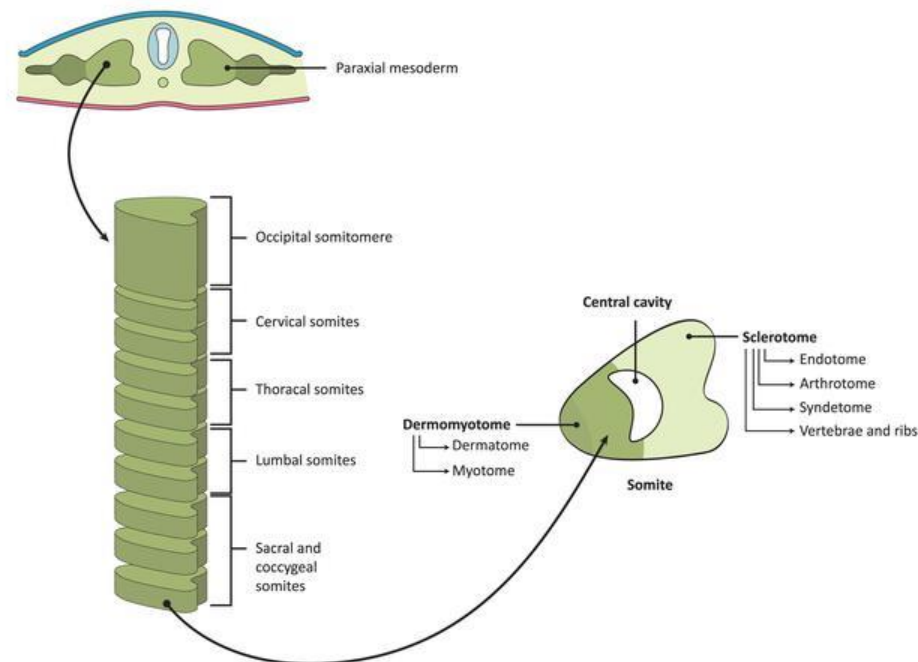


INTRODUCTION

- Muscles develop from **intraembryonic mesoderm** except muscles of iris (ectoderm of optic cup)
- Muscle fibers develops from embryonic cells=myoblasts (derived from mesenchyme)
- **Skeletal muscle** derived from **paraxial mesoderm** which forms somites from the occipital to the sacral regions and somitomeres in the head
- **Cardiac muscle** and **Smooth muscle** of GIT, CVS, and Respiratory system differentiates from **splanchnic mesoderm**

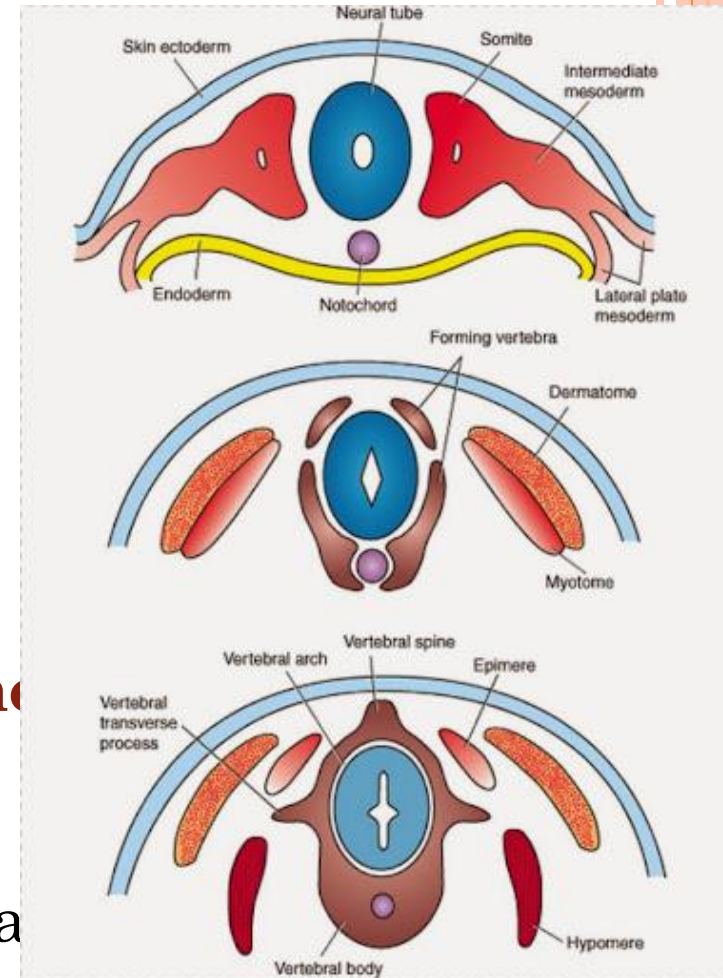
SKELETAL MUSCULATURE

- Develop from **paraxial mesoderm**
 - Trunk from myotome
 - Limb from limb buds
 - Face & neck from **pharyngeal arches**
- **Somites** and **somitomeres** form the musculature of the axial skeleton, body wall, limbs and head



SKELETAL MUSCULATURE

- From the occipital region caudally, somites formed
- Each somite differentiates into two distinct zones: **sclerotome** and **dermomyotome**
- The **sclerotome** forms **bones of axial skeleton**. The **dermomyotome** consists of two components: a deeper part called **myotome** and a superficial part called **dermatome**.
- The **myotome** forms **muscular tissue** whereas the **dermatome** forms **dermis of skin**



DEVELOPMENT OF SKELETAL MUSCLES IN DIFFERENT REGIONS OF THE BODY

- In the human body, the skeletal muscles are divided into following groups:
 - Muscles of the body wall (trunk)
 - Muscles of head and neck
 - Muscles of limbs

Muscles of the body wall (trunk)

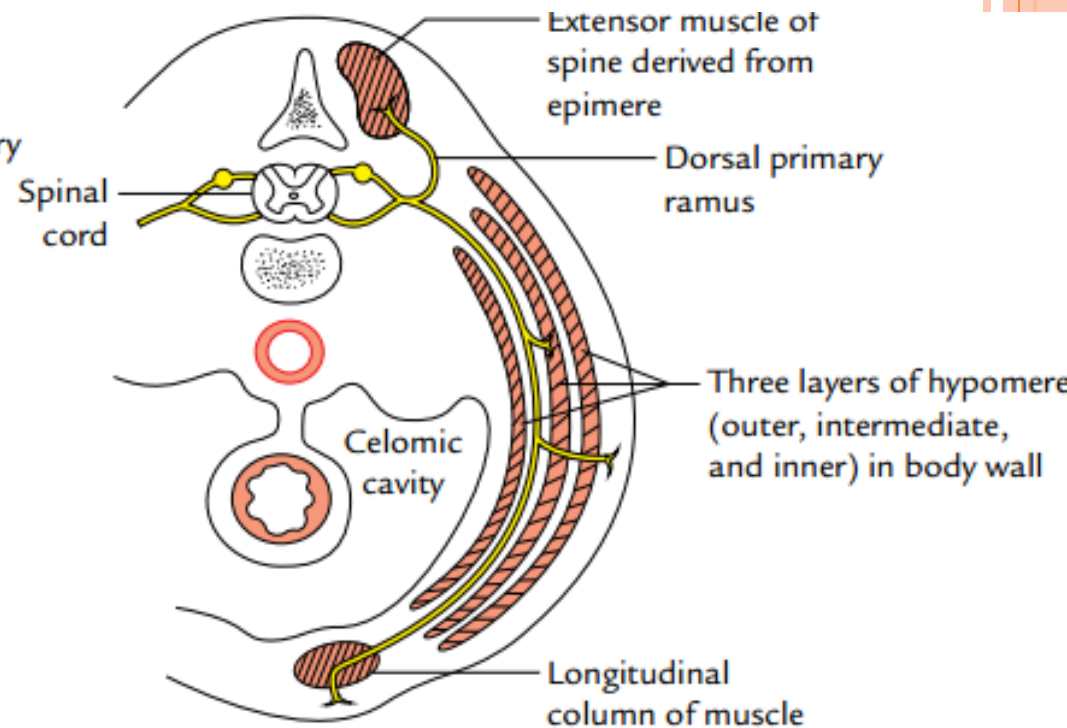
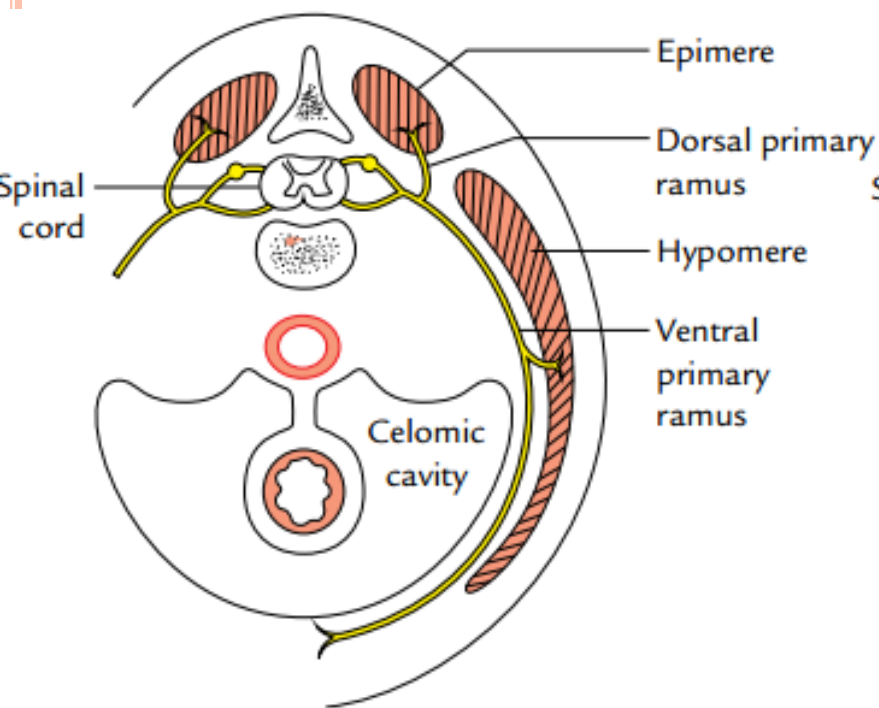
- Muscles of the body wall are derived from **myotomes** (somites) of the trunk region.
- Each myotome divides into two parts:
 - a smaller dorsal part called **epimere**
 - a larger ventral part called **hypomere**
- The muscles derived from **epimere** and **hypomere** are supplied by **dorsal ramus** and **ventral ramus** of spinal nerve respectively.

Muscles of the body wall (trunk)...

- Myoblasts of the epimeres form the **extensor muscles** of the vertebral column, (e.g., erector spinae) and those of the hypomeres give rise to muscles of the **limbs and body wall**
- The hypomeres of myotomes extend ventrolaterally and form following **muscles of the body wall**.
 1. They form three layers of **muscles in the thorax**, external intercostal, internal intercostal, and innermost intercostal or transversus thoracis.
 2. In the **abdomen** also they form, external oblique, internal oblique, and transversus abdominis
 3. In the **neck**, they form longus colli, longus capitis, and scalene muscles.

Muscles of the body wall (trunk)...

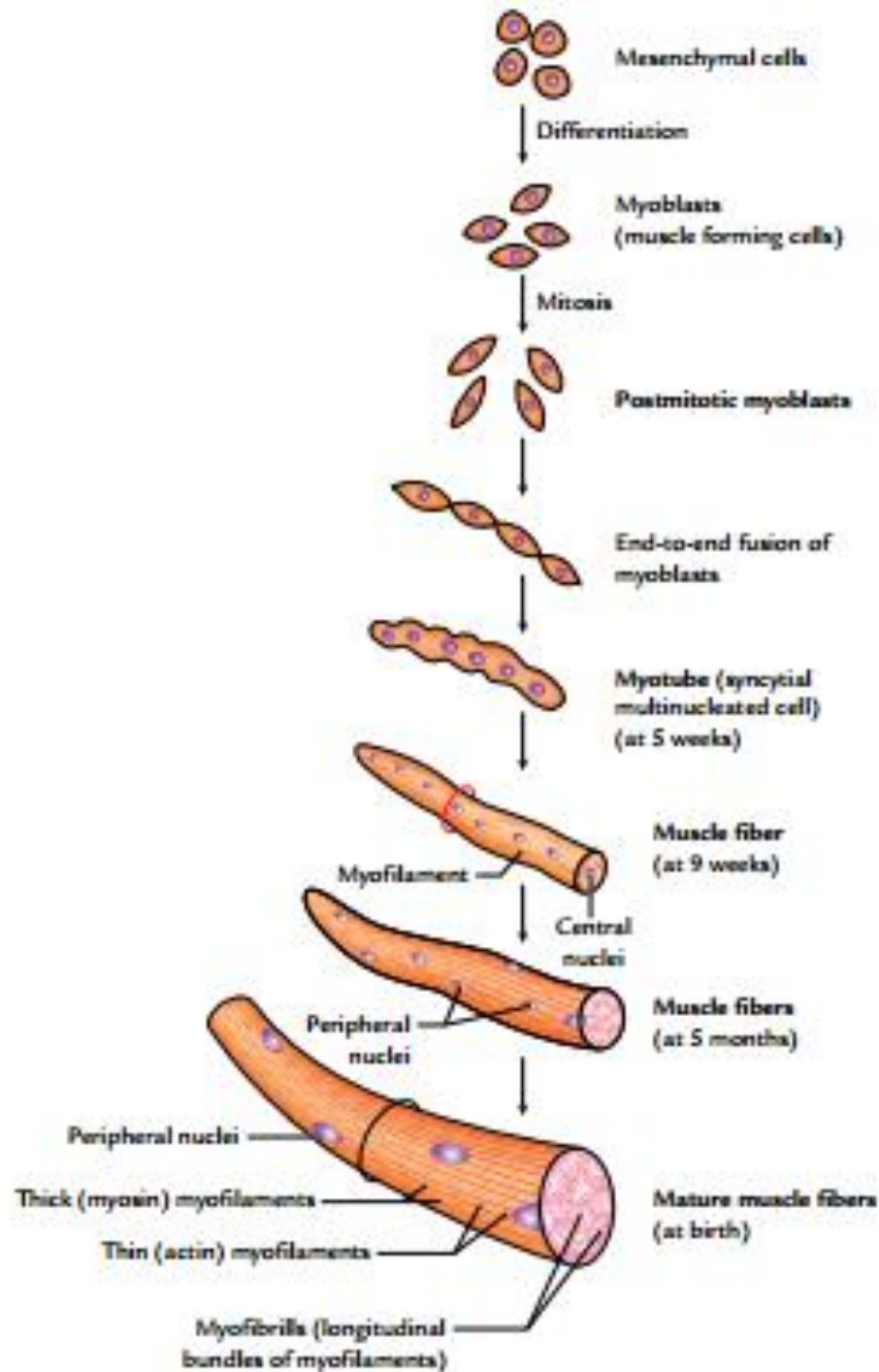
- Myoblasts from the hypoblast of lumbar segments form the **quadratus lumborum muscle**, and those from sacral and coccygeal regions form the **pelvic diaphragm and striated muscles of the anus**
- In addition to the three ventrolateral muscle layers, a ventral **longitudinal** column arises at the **ventral tip** of the hypomeres
 - **Rectus abdominis muscle** in the abdominal region, **infrahyoid musculature** in the cervical region and **sternalis muscle** in the thoracic region which normally disappears



Subdivision of Myotome into **Epimere** and **Hypomere**

HISTOGENESIS

- The skeletal muscles develop from mesenchyme derived from myotome.
- The mesenchymal cells differentiate into primordial muscle cells called **myoblasts**.
- The myoblasts fuse with each other end-to-end to form an elongated multinucleated cylindrical cell called **myotube**.
- The myotubes synthesize **actin**, **myosin**, and **other muscle proteins**.
- These proteins aggregate to form **myofilaments** and **myofibrils**.
- Now, the myotubes are called **muscle fibers**.
- A number of these muscle fibers are bound together by connective tissue to form **individual muscles**



MUSCLES OF HEAD AND NECK

Extraocular muscles (extrinsic muscles) of eyeball

- They develop from three **preotic myotomes** that are arranged around developing eye.
- These muscles are innervated by IIIrd, IVth, and VIth cranial nerves

Muscles of tongue:

- They develop from **occipital somites**
- The **hypoglossal nerves** supply all the intrinsic and extrinsic muscles of the tongue except palatoglossus.

MUSCLES OF HEAD AND NECK...

Muscles of pharyngeal arches:

- They develop from mesoderm of pharyngeal arches
- The muscles derived from mesoderm of pharyngeal arches are **muscles of mastication, facial expression, pharynx, and larynx.**
- These muscles are innervated by **nerves of the respective pharyngeal arches.**

Muscles of pharyngeal arches...

Muscles of First Arch

- **Muscles of mastication** (temporalis, masseter, lateral pterygoid, and medial pterygoid),
- Anterior belly of digastric, mylohyoid, tensor tympani, and tensor palati.
- All these muscles are supplied by **mandibular nerve**.

Muscles of Second Arch

- **Muscles of the facial expression**, posterior belly of digastric, stylohyoid, and stapedius.
- All these muscles are supplied by **facial nerve**.

Muscles of pharyngeal arches...

Muscles of Third Arch

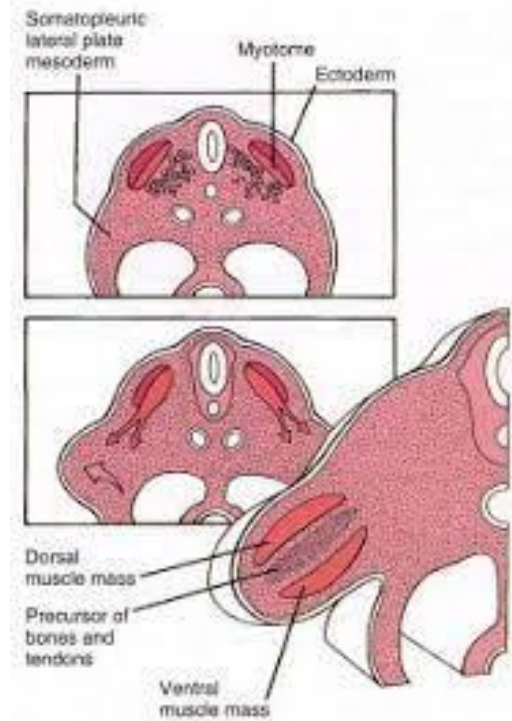
- The only muscle is **stylopharyngeus**.
- It is supplied by **glossopharyngeal nerve**.

Muscles of Fourth and Sixth Arches

- **From 4th** Cricothyroid, levator veli palati, and constrictors of the pharynx.
- They are supplied by **superior laryngeal nerve**.
- **From 6th** arch are **intrinsic muscles of the larynx**.
- They are supplied by **recurrent laryngeal nerve**

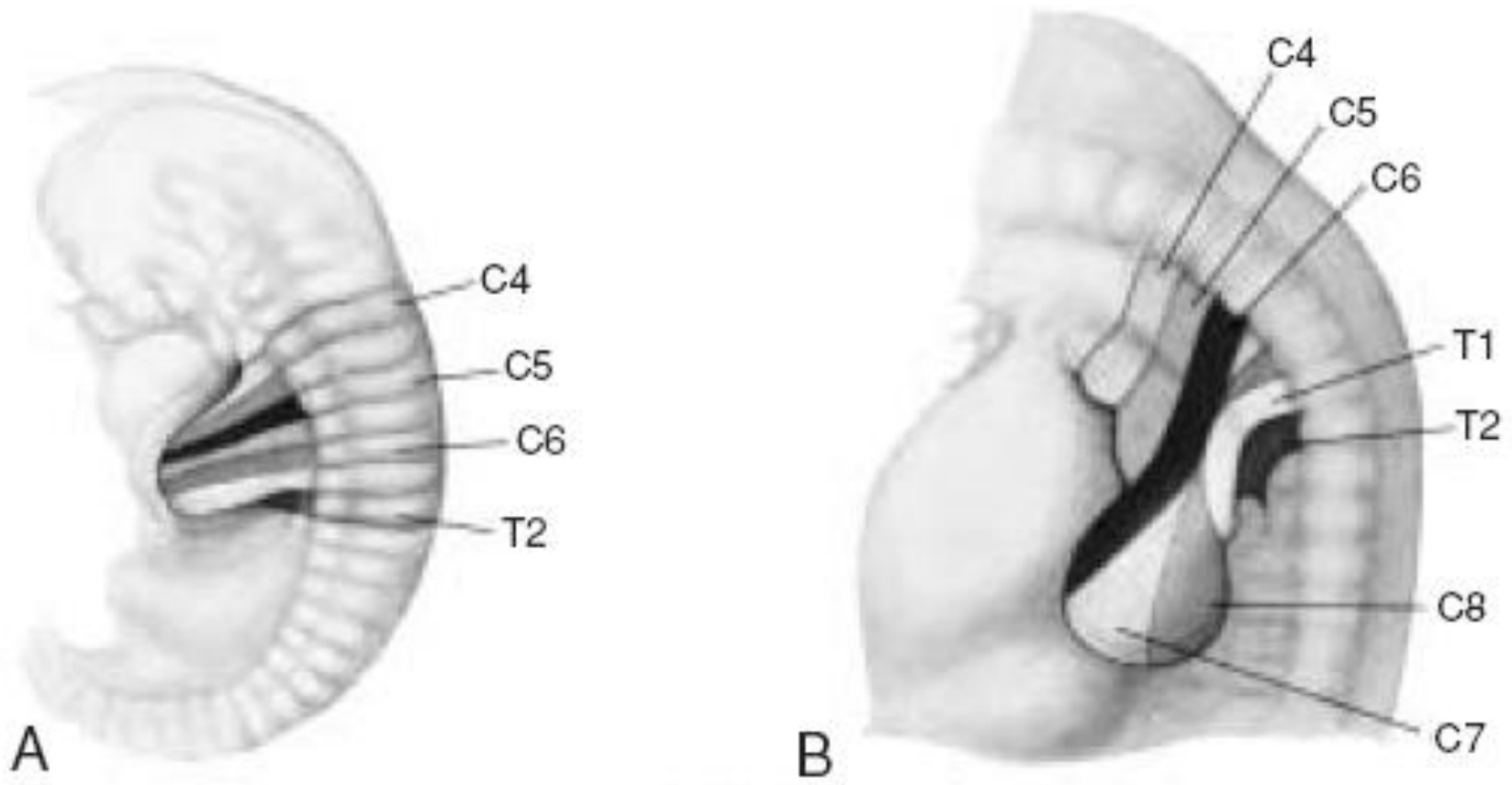
MUSCLES OF LIMBS

- At 5th week of development, the mesenchyme is derived from mesoderm migrate into base of limb buds and forms **anterior** and **posterior condensations**.
- The **anterior condensation**
 - On upper limb gives rise to **flexor** and **pronator muscles** and
 - On lower limb **extensor** and **adductor** muscles.
- The **posterior condensation**
 - On upper limb gives rise to **extensor** and **supinator muscles** and
 - On lower limb **flexor** and **abductor** muscles.

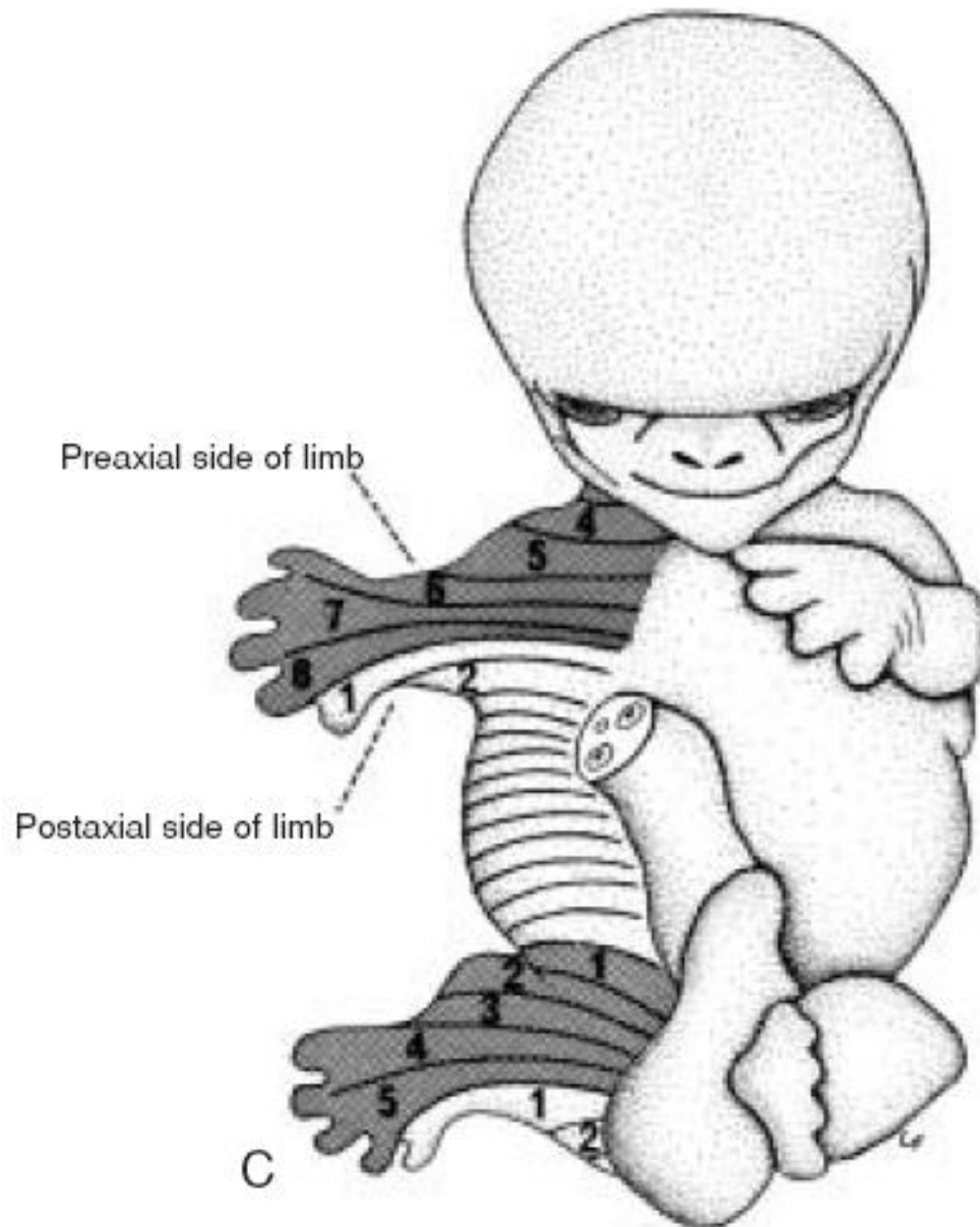


LIMB BUDS WITH THEIR SEGMENTS OF ORIGIN INDICATED.

A. UPPER LIMB BUD AT 5 WEEKS. **B.** UPPER LIMB BUD AT 6 WEEKS.



C. LIMB BUDS AT 7 WEEKS



CONGENITAL ANOMALIES

○ Absence of muscles

- Partial or complete
- E.g. total or partial absence of pectoralis major muscle (**Poland anomaly**), palmaris longus, serratus anterior, and quadratus femoris

○ Congenital torticollis: sternocleidomastoid muscle injury at birth

DEVELOPMENT OF SMOOTH

- Smooth muscle in the wall of the **digestive** and **respiratory** tract is derived from **splanchnic mesoderm** surrounding the endoderm of these structures
- **Vascular smooth muscle** (blood and lymphatic vessels) differentiates from mesoderm adjacent to vascular endothelium
- Muscles of **iris** (spincter and dilator pupillae), **arrector pili** muscles of skin, and myoepithelial cells of sweat and mammary glands develop from mesenchymal cells derived from **ectoderm**.

CARDIAC MUSCLE

- Cardiac muscle develops from splanchnic mesoderm surrounding the endothelial heart tube
- Primitive myocardium is recognized by 4th week
- Development
 - Arise by differentiation of single cell
 - Myofibrils develop as in skeletal muscle but myoblasts do not fuse
 - Myoblasts adhere to one another by special attachments that later develop into **intercalated discs**
 - Later, network of cardiac muscle with increased size and few myofibrils appear and give rise to purkinje fibers of the conducting system

The
End

